

Assignment 2 – Event Handling [Part 1]

Goal

To put into practice the event driven development principles and techniques covered in class, including function pointers, event handlers and callbacks. The goal of this assignment is to implement an event handling system for game development. This framework is a fundamental building block of modern game architectures, so it is important that all students gain some experience with its internal design and implementation.

Description

The event handling system implemented will support the two different approaches discussed in class: an event system based on callback functions and a generic event handler using templates. These systems are independent each other. For compatibility the name of the classes and some methods have been changed. The objective as the previous assignments is to get familiar with their implementations and create the foundations to apply event handling on a videogame. In this sense, it is necessary to full fill the following requirements:

1. Base Event class (event.hh)

You have to implement an empty abstract class **Event**. All Events must inherit from this class.

2. Event system based on callbacks (event_callbacks.hh):

Callback function:

You have to implement a function pointer to handle all events. The name of the callback function datatype must be **CallbackFunction**.

EventListener: Object containing the who and the how

Attributes	Description
Context	Object interested in the event
CallbackFunction	Callback function to be executed once the event is raised

EventDispatcher: Class that registers the objects listening the event and their callback function. It stores and notifies of an event to the listeners.

Attributes	Description
Listeners Subscribed	Array or container of listeners subscribed to the events
Methods	Description
subscribe	Allows listener objects to be registered with his static callback method
getListenerAt	Allows to get a listener depending on their position in the array
dispatchEvent	Notifies all the listener that an event has happened

3. Generic event handling (event_handlers.hh):

IEventHandler: Interface that all Event handlers must implement

Methods	Description
call	Method that receives and handles the event

EventHandler: Generic event handler class implemented using templates, requires to type names: **T** for the type of the object interested in the event and **E** for the type of the specific event.

Attributes	Description
Object T	Object of type T interested in the Event
ActionMethod	Function pointer to the method of the instance handling the event. It should receive as parameter an event of type E.
Methods	Description
constructor	Constructor of the event handler class, it should receive a pointer to an object T interested in the event and the function pointer to the method in the class handling the event (ActionMethod)
call	Implement the call method from the parent interface. it should execute the function pointer ActionMethod in the object of type T, passing an event of type E as parameter.

EventManager: Registers all the event handlers listening to the event and notifies them.

Attributes	Description
Listeners Subscribed	Array or container of event handlers subscribed to the events
Methods	Description
registerEventHandler	Allows event handlers to be registered with his static callback method
getEventHandlerAt	Allows to get an event handler depending on their position in the array
notifyEvent	Notifies all the event handlers that an event has happened

The primary objective of this first part is to ensure that all tests compile and pass successfully. Code quality, documentation, and overall organization will also be evaluated. Students should not modify any files related to the tests; only the files specified for the assignment can be changed. Upon completion, students should have implemented a usable framework capable of handling events.

What to do:

The code explicitly requires the following files, which the student should create and submit:

- event.hh, event_callbacks.hh, event_callbacks.cc, event_handlers.hh and event_handlers.cc

Submission Guidelines

Submit all the files as a zip folder on Moodle with the following naming conventions:

cs225fa26_{username}_2.1.zip